ECSE 331

PSPICE Assignment No. 3

MOSFET DC Characteristics in PSPICE[1]

Objective

Investigate the i-v characteristics of MOSFET through PSPICE simulations.

Practical Information for the Student:

In order to revisit the basics of simulation in the PSPICE and to have an overview of the basic steps for NMOS simulation, students are encouraged to go through tutorial available here[2]:
http://www1bpt.bridgeport.edu/~thapa/eleg458/tut/pspice/PSPICE_MOS_TUT5.pdf

In order to get familiarized with tools available for plots in PSPICE which may be helpful in writing this report, students are encouraged to go through tutorial available here:
https://drive.google.com/open?id=1ufWwsecdJN9Yh0l8aUi8TOCsurQsMx4i

Write-Up Requirements:

A good laboratory report should contain a brief description of what the experiment was about, including circuit diagrams, and what you did, your data, your results, and anything else called for in the assignment, such as questions inserted in the laboratory. Answers to these questions require observations that need to be made at the time you do the experiment. The laboratory report should be written using the IEEE paper style consisting of a double-column single-space format, and must adhere to the following:

1. Title page - Title of the assignment/project, authors' name, and course name.
2. Abstract - Abstract of the assignment/project report.
3. Introduction
4. Main body of the assignment/project report including figures.
6. Conclusions
7. References
8. Appendices

[1] The choice of PSPICE as the simulation software is not strict. The student may use any other simulation tool he/she is comfortable with.

[2] When you start Capture.exe in Step 1 of the above tutorial, right click on Capture.exe and select “Run as administrator”.

Procedure:

I. MOSFET i_D-v_{DS} Characteristics

Construct the circuit shown in Fig. 1 in PSPICE Capture.

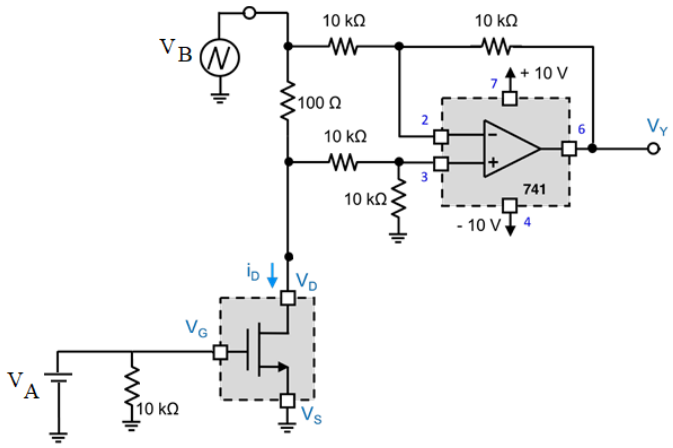


Fig. 1. Set-up for measuring MOSFET i-v characteristics

For the op-amp, use the component "uA741" from the PSPICE parts library as shown in Fig. 2. The OS1 and OS2 pins of uA741 should be left unconnected. The DC voltages of -10V and +10V on pin 4 and 7 respectively can be implemented using two simple DC sources (Vdc in PSPICE).

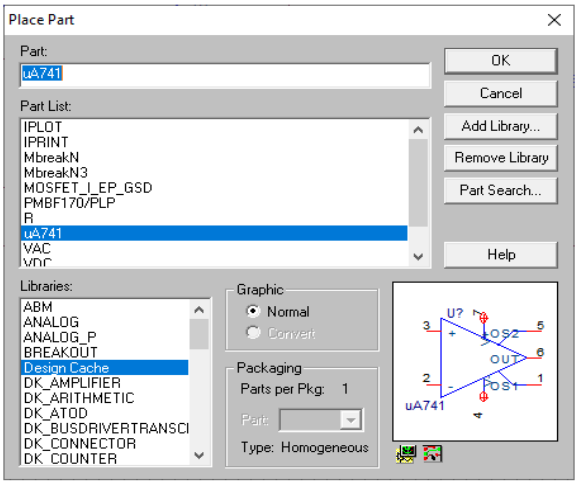


Fig. 2. uA741

For the NMOS, use the component “MbreakN3” as shown in Fig. 3.

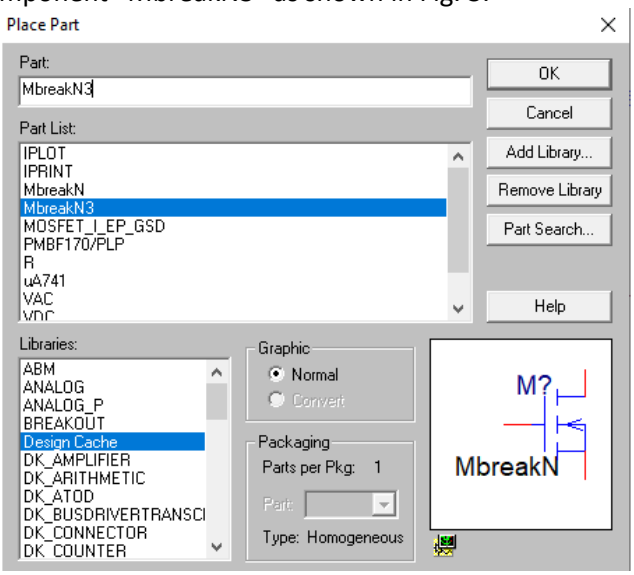


Fig. 3. MbreakN3

For MbreakN3, set W= 10µm, L = 1µm and use the following description:

```
* 1 um Level 3 models
*
* Don't forget the .options scale=1u if using an Lmin of 1
* 1<Ldrawn<200 10<Wdrawn<10000 Vdd=5V
.MODEL Mbreakn NMOS LEVEL = 3
+ TOX = 200E-10 NSUB = 1E17 GAMMA = 0.5
+ PHI = 0.7 VTO = 0.8 DELTA = 3.0
+ UO = 650 ETA = 3.0E-6 THETA = 0.1
+ KP = 120E-6 VMAX = 1E5 KAPPA = 0.3
+ RSH = 0 NFS = 1E12 TPG = 1
+ XJ = 500E-9 LD = 100E-9
+ CGDO = 200E-12 CGSO = 200E-12 CGBO = 1E-10
+ CJ = 400E-6 PB = 1 MJ = 0.5
+ CJSW = 300E-12 MJSW = 0.5
```

For V_B in Fig. 1, generate a saw-tooth waveform over a 0 V to +10 V signal range and a period of 1ms using the component “VPLUSE” with the following characteristics: V1=0, V2=10, TD=0, TR=0.49e-3, TF=0.49e-3, PW=1e-5, PER=1e-3.

V_A is a simple DC source (Vdc in PSPICE). Beginning with the gate voltage V_A set to 0 V, simulate the circuit and **plot v_{DS} vs time and i_D vs time on separate figures**. You should observe that the drain current i_D remains zero or negligible irrespective of any change in drain-source voltage v_{DS}.

Next, increase the gate voltage in steps of 1V until you get i_D>0.5mA. Consider this value of gate voltage as the NMOS threshold voltage. **Report the value of the threshold voltage. Plot the voltage V_Y (obtained at the output of the op-amp as shown in Fig. 1) divided by 100 vs time. Also, plot the current through the 100Ω resistance vs time. Compare the two plots and explain what you observe.**

Next, increase the gate voltage in steps of 1V until the gate voltage reaches 5V, and obtain plots of the i_D-v_{DS} characteristics for each value of the gate voltage. **In your report, combine all the curve traces into a**

single plot and explain the overall behavior based on your theoretical knowledge of NMOS[3]. What is the effective Early voltage of your device?

II. MOSFET Transconductance

Set V_A to 3V and the V_B to 5V. This will require that the saw-tooth signal source V_B be replaced with a DC voltage source V_B. Once complete, measure the gate voltage and drain current, and record these results. Next, adjust the gate voltage by approximately 100 mV. Again, measure the gate voltage and drain current, and record these results. **Compute the transconductance g_m of your device** using the following formula:

$$g_m = \frac{\Delta i_D}{\Delta v_{GS}}$$

III. MOSFET Temperature Effects

Reconstruct V_B as the sawtooth waveform signal and set to a 0- 10 V sweep as in Part l. Set the gate voltage to 3 V. Run the simulation at 5 different temperatures: 10^0 C, 20^0 C, 27^0 C, 35^0 C and 50^0 C by editing the simulation profile as shown in Fig. 4. **Plot the i_D-v_{DS} characteristics at those temperatures. How do they compare? Provide an explanation with supporting theory.**

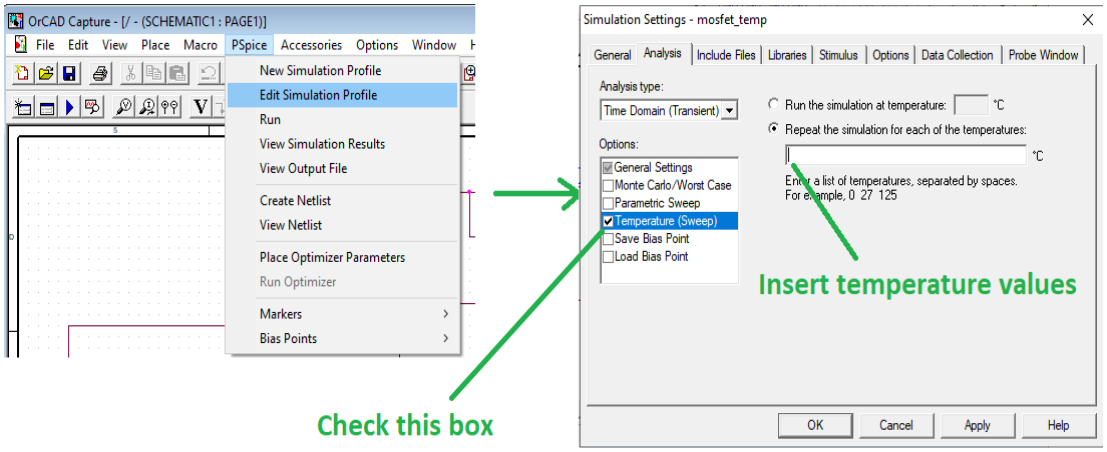


Fig. 4. Temperature sweep in PSPICE

This concludes this lab.

[3] An easy way to combine curves generated individually in PSPICE into one plot is to transfer the data of each plot to Excel/Matlab etc. and generate the combined plot there.